BCH 4102: Tissue Biochemistry

Lecture I

Biochemistry of kidney

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❖ Five Basic Functions are:

- Production of adequate amount of normal urine and Retention of vital substances.
- Regulation of body water, electrolyte and acid –base balance (***isovolumia, isoionia, isohydria***)
- Excretion of nitrogen waste (urea, ammonia, uric acid) and other waste.
- Participation in biosynthesis (creatine, carnitine, glucose, AA) (**outline the biosynthetic pathway**). it contribute about 50% of the blood glucose via gluconeogenesis in starvation
- Endocrine functions.

- Kidneys are only 0.5 - 1% by body mass, receive about 25% of cardiac output and consumes 10% of the daily oxygen.
- Significant amount of tubular glucose move into the kidney cells by the Na^+-glu cotransporter
- This cotransporter is indirectly powered by Na^+-K^+ ATPase that maintained the Na^+-K^+ gradient.

❖ Functional parts (Overview)

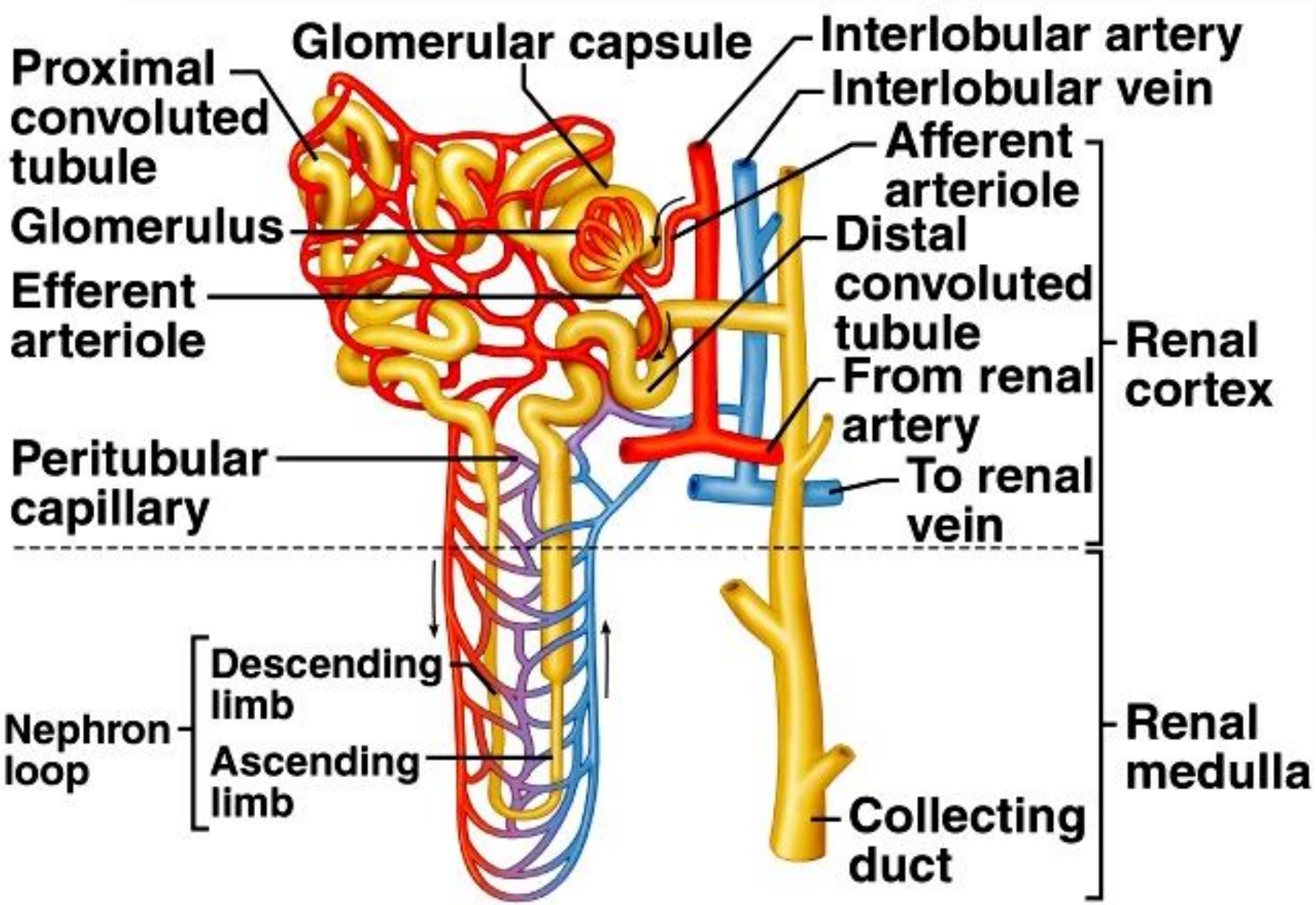
- **The functional unit, nephron has 3 functional parts viz :-**

I. Renal corpuscle:

- Glomerular capillaries
- Bowman's capsule (parietal and visceral layer)

II. Renal tubules – tubular cells

III. Interstitial cells


Glomerular capsule
Interlobular artery
Proximal convoluted tubule
Interlobular vein
Afferent arteriole
Glomerulus
Distal convoluted tubule
Efferent arteriole
Renal cortex
From renal artery
Peritubular capillary
To renal vein
Descending limb
Nephron loop
Ascending limb
Renal medulla
Collecting duct

❖ Blood supply/circulation (Overview)

- ➢ *Blood supply, circulate and leave the nephron by:*
- ✓ Afferent arteriole
- ✓ Glomerulus
- ✓ Efferent arteriole
- ✓ Peritubular capillaries or arteriolae
- ✓ vasa recta

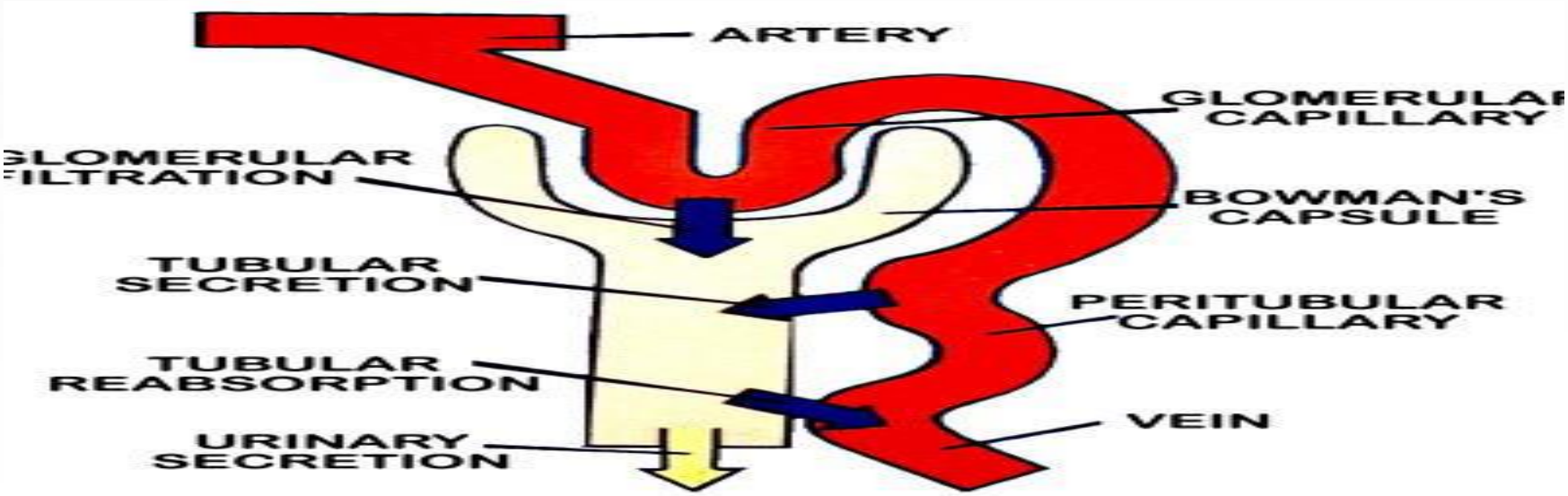
❖ Renal tubules (Overview)

Tubules of a nephron are:-

- Bowman's capsule (parietal and visceral layer)
- Proximal convoluted tubule
- Loop of Henle - thick and thin each with descending and ascending limbs
- Distal convoluted tubule
- Collecting duct

❖ Mechanism of urine production

- Glomerular filtration
- Tubular Selective (active) and passive reabsorption
- Tubular Secretion
- Urinary excretion rate = Filtration rate – (Reabsorption rate + Secretion rate)



❖ Glomerular filtration

It involves straining plasma through a molecular sieve composed of:

- Capillary endothelium (glomerular capillaries)
- Basal membrane: negatively charge glycoproteins (collagen IV)
- Podocyte processes (Bowman's capsule)

- **Events at Bowman's capsule:**

Blood passes the juxtaglomerular apparatus into the glomerulus through the afferent arteriole and leave the glomerulus through the efferent arteriole, while the filtrate leave the bownan's capsule through the proximal tubule.

- ***Filtration Mechanism*** :

- ✓ Fluids pass out of capillaries through the podocytes of bowman's capsule into the capsule lumen.
- ✓ 20% of flowing plasma is filtrated (2ml/second) about 160 – 180 L daily.
- ✓ plasma is filtered about 60 times per day

- ✓ The primary urine within Bowman's capsule (~ 150ml), is identical to plasma except:-
  - traces of α_2, and β_2 microglobulin, lysozyme that subsequently reabsorbed in tubules.

- **Pathology – *glomerular filtration impairments***

✓ Proteinuria (PU) = presence of protein in final urine (>0.15g/24hrs urine) could be due to:-

- mechanical (non-selective PU)
- charge loss (selective PU)

❖ Tubular reabsorption and secretion

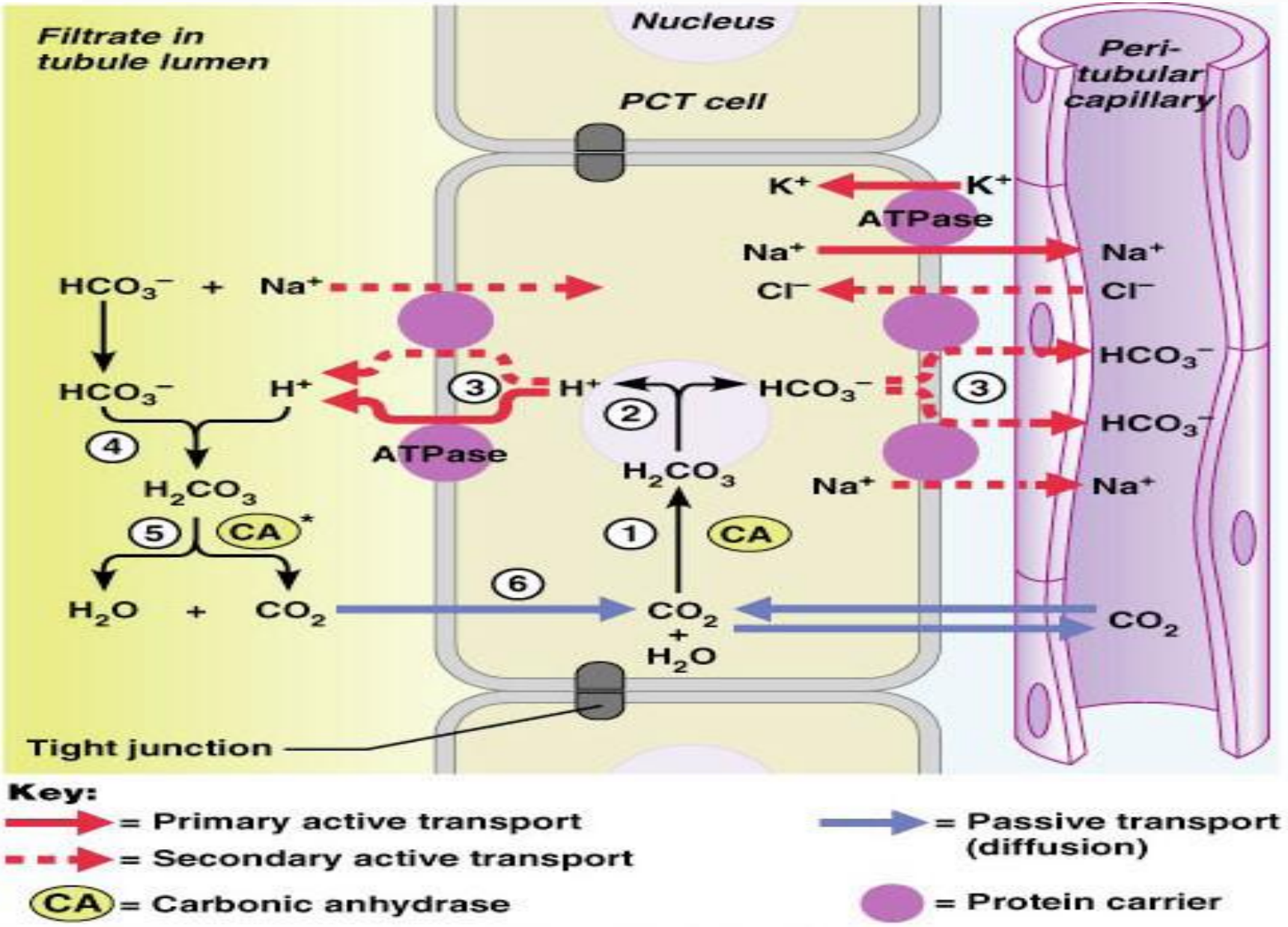
- ✓ Renal tubules are sophisticated tubules that:
  - analyze urine
  - excrete or save back substances according to body needs.
- ✓ It involves active transport (ATP) and hormonal regulation (ADH, ALDESTERONE and PTH).

I - Proximal tubule

- ✓ It is hormone – independent site of obligatory reabsorption of;
  - Most Na^+, Cl^-, K^+
  - ~ 70% of water

(*actively reabsorbed **Na**$^+$ is followed by **Cl**$^-$ and **water***)

  - All filtered glucose and amino acids (*secondary active transport of **glc** and **Aas***)
  - Bicarbonate: the H_2CO_3 dissociates to CO_2 and H_2O, the former diffuses into tubular cells to trigger more hydrogen ion secretion (HOW?).


Filtrate in tubule lumen
Nucleus
PCT cell
Peri-tubular capillary
K+
K+
ATPase
Na+
Na+
HCO3− + Na+
Cl−
Cl−
HCO3−
HCO3− H+
3
H+
HCO3−
3
HCO3−
4
2
ATPase
H2CO3
Na+
Na+
H2CO3
5
CA*
1
CA
6
H2O + CO2
CO2 + H2O
CO2
Tight junction
Key:
= Primary active transport
= Secondary active transport
CA = Carbonic anhydrase
= Passive transport (diffusion)
= Protein carrier

II - Loop of Henle

- Further re-sorption of 20% of filtrated volume
- Induction of hyperosmolar condition in the kidney medulla
- Counter-current multiplication system:
  - Descending limb is freely permeable for water
  - Ascending limb has $Na^{+,}$ K^{+}, $2Cl^{-}$ co-transporter

Assignment: describe counter – current mechanism

III Distal tubule

- Aldosterone-dependent Na^+ reabsorption and K^+ excretion
- Further reabsorption of 0 - 5% filtered water
- Aldosterone:
  - increases Na^+ and fluid conserving in the body
  - increases K^+ excretion in the distal tubule acts by inducing Na^+ K^+ ATPase of basolateral membrane

IV Collecting tubule

- Determines urine final volume and osmolality
- ADH increases tubule wall permeability for water, thus concentrating urine:
- In the presence of ADH water is passively moved to hyperosmotic intersticium

(hyperosmotic intersticium is maintained by the Henle´s loop) – hyperosmotic urine is produced

- Otherwise urine pass the kidney medulla unchanged (hypoosmotic urine is produced)

Constituents of the urine

- Organic constituents
- Inorganic constituents

Organic constituents

	grams/day
urea	20 – 30
creatinin	1 – 1,5
uric acid	0,3 – 2,0
hippurate	0,15
glucose	< 0,16
ketone bodies	< 3
aminoacids	< 1 – 3
proteins	< 0,15

Inorganic constituents

	mmol/day
Cl^-	120 – 240
Na^+	100 – 150
K^+	60 – 80
SO_4^{2-}	30 – 60
NH_4^+	30 – 50
HPO_4^{2-}	10 – 40
Ca^{2+}	4 – 11
Mg^{2+}	3 – 6

❖ Proton secretion

- The tubule cells absorb CO_2 from the blood and then hydrate it to carbonic acid (carbonate dehydratase).
- Carbonic acid then dissociates to HCO_3^- and H^+.
- H^+ pass in the urine by an ATP- driven membrane-localised transport system, while HCO_3^- returns to the blood.

❖ Ammonia excretion

- Ammonia is produced enzymatically from glutamine and other AA.
- NH_3 combines with secreted H^+ ions to form a nondiffusible ammonium ion (NH_4^+) which is excreted in the urine.
- Ammonia production is increased by a severe metabolic acidosis.

❖ Recycling of the calcium and phosphate ions

- Parathormone (parathyrin)
- Calcitonin
- Calcitriol

➢ Parathormone PTH (parathyrin)

- produced by parathyroid gland to:
- stimulates the absorption of calcium in the kidney
- at the same time inhibits the resorption of phosphate
- Hence elevated plasma PTH levels leads to:
- an increase in plasma calcium level
- decline in plasma phosphate level

➢ Calcitonin

- A peptide produced in cells of the thyroid gland
- Inhibits the reabsorption of both calcium and phosphate
- The result is an overal decline in the plasma level of either ion
- With to respect to calcium reabsorption, calcitonin is an antagonist of PTH

➢ Calcitriol

- Steroid hormone, formed in the kidneys
- It has stimulatory effect on the reabsorption of both calcium and phosphate ions

➢ Reabsorption of sodium ions

- Aldosterone (steroid hormone) stimulates Na^+ retention
- Atrial natriuretic peptide (ANP), a hormone from the atrium of the heart inhibits retention of Na^+.
- Both hormones probably affect Na+/K+ - ATPase

➢ Atrial Natriuretic Peptide (ANP)

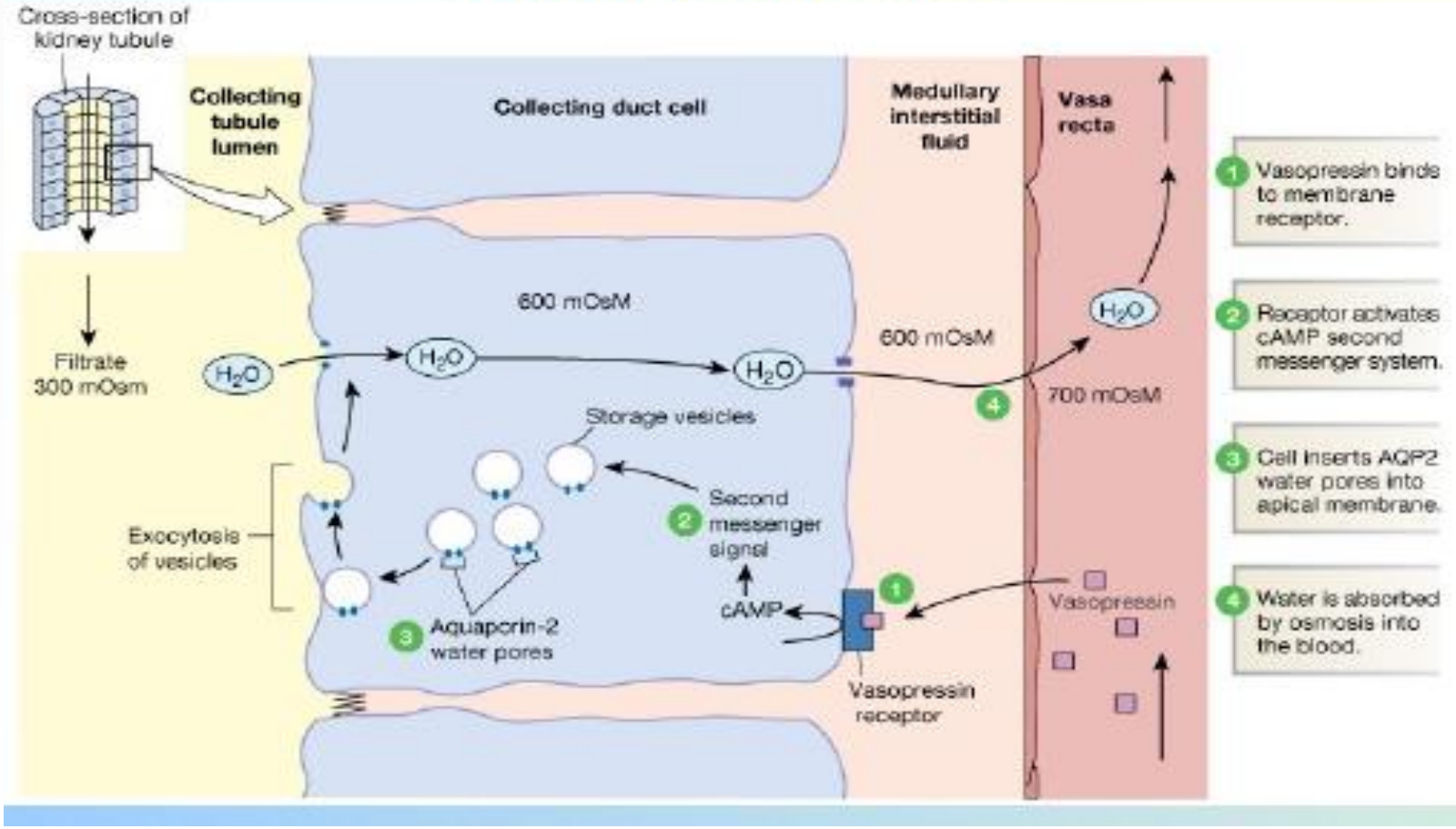
- ANP is released when arterial pressure is increased (in heart failure) or fluid overload.

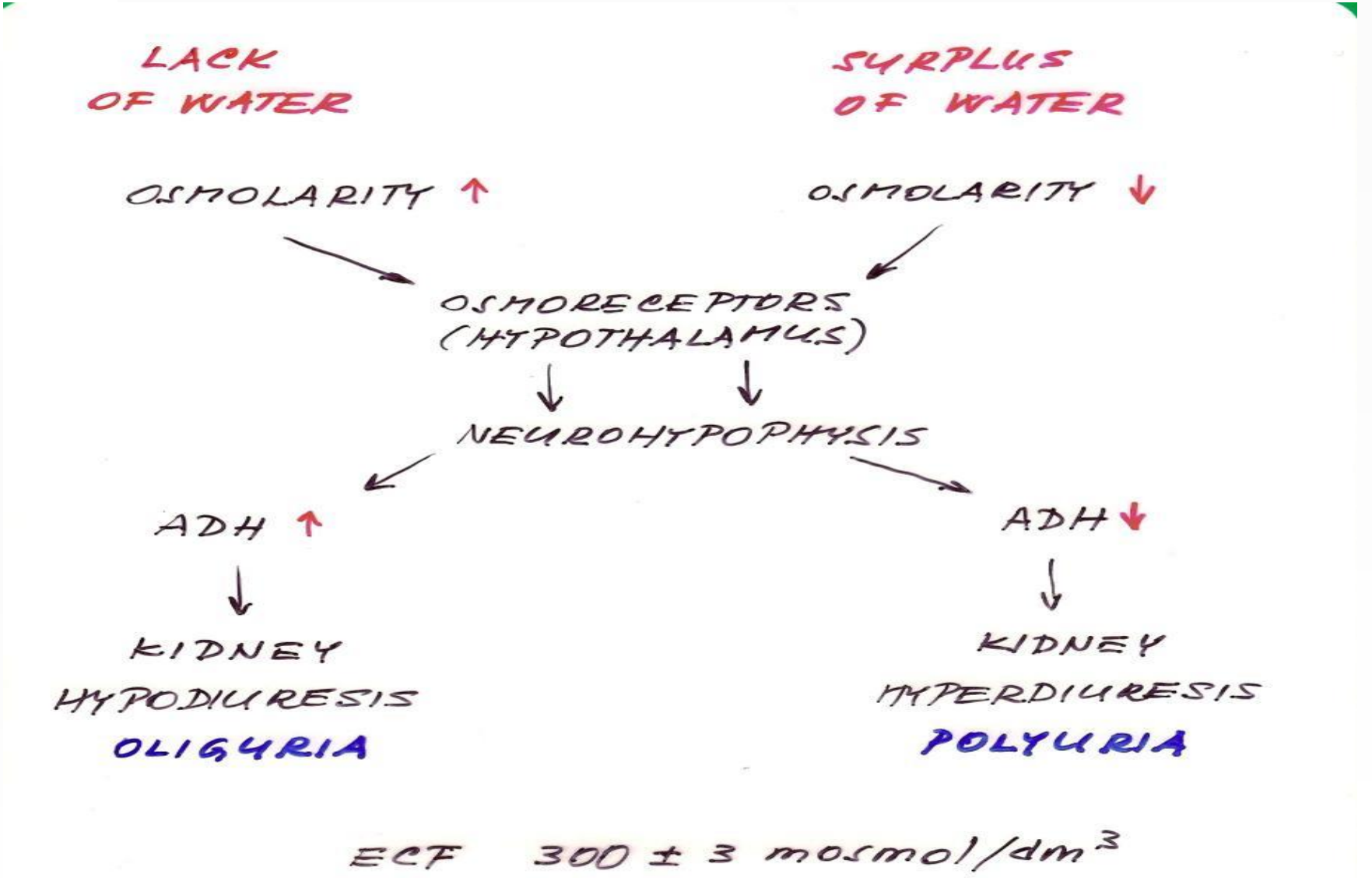
- It promotes loss of Na^+ and Cl^- ions and water chiefly by increasing GFR.

Antidiuretic Hormon (ADH)

- ADH increases the water permeability of the distal tubule and collecting duct, thus increasing the concentration of urine.
- In contrast when secretion of ADH is inhibited, it allows dilute urine to be formed. This occures mainly, when plasma sodium concentration falls such as following drinking large quantities of water.
- This fall is detected by osmoreceptors in the hypothalamus.

ADH function




LACK OF WATER
SURPLUS OF WATER
OSMOLARITY ↑
OSMOLARITY ↓
OSMORECEPTORS (HYPOTHALAMUS)
NEUROHYPOPHYSIS
ADH ↑
ADH ↓
KIDNEY
HYPODIURESIS
OLIGURIA
KIDNEY
HYPERDIURESIS
POLYURIA
ECF 300 ± 3 mosmol/dm³

Gluconeogenesis

- Chief substrate is glutamine
- Other substrates are AA, lactose, glycerol or fructose (obtained from blood plasma)
- The ammonia derived from this AA serve to buffer the pH of the urine.
- Gluconeogenesis is induced by cortisol

Glucose recycling

- Is an energy dependent process
- Is independent of gluconeogenesis
- Glucose uptake occurs as compulsatory cotransport with Na^+ ions
- It is driven by the concentration gradient of Na^+ between the urine and interior of the cells (secondary active transport)

Hormones of the kidney

- Erythropoietin
- Calcitriol
- Angiotensin

Erythropoietin

- A peptide hormon that is formed predominantly by the kidney (also by the liver)
- controls differentiation of bone marrow stem cells
- The release is stimulated by hypoxia (low pO2)

Calcitriol

- 1-alpha, 25-dihydroxycholecalciferol is a steroid-related hormon involved in calcium homeostasis.
- It is formed in the liver from calcidiol by hydroxylation at C-1
- The activity of hydroxylase (calcidiol-1-monooxygenase) is regulated by the hormone parathyrin (parathormone).

Renin – angiotensin system

- Renin is an enzyme which converts the plasma protein angiotensinogen to angiotensin I.
- Angiotensin converting enzyme (ACE) which is formed in the lungs converts angiotensin I to angiotensin II which causes vasoconstriction and an increase in blood pressure.
- Angiotensin II also stimulates the aldosterone production (water and sodium retention which together increase blood volume).

- Therefore renin via angiotensin II, when there is fall in intravascular volume and dehydration, leads to:

  - Constriction of the efferent arteriole to maintain GFR, by increasing the filtration pressure in the glomerules.
  - Release of aldosterone.
  - Increased release of ADH.
  - Thirst
  - The opposite occurs when in fluid overload.

The hormones interact when blood loss or dehydration occurs to maintain intravascular volume

